

MENDSPEECH REVISED LEARNING SYSTEM

MendSpeech PDF Set Index

Recommended reading order for the revised complete plan.

The complete daily plan contains all 56 days. The weekly PDFs are easier to use while executing the project.

Files

Document	File	Use it for
MendSpeech Project Blueprint	MendSpeech_Project_Blueprint.pdf	Architecture, research questions, metrics, benchmark package, and definition of done.
MendSpeech 8 Week Master Roadmap	MendSpeech_8_Week_Master_Roadmap.pdf	Timeline, phases, workload, hardware, research questions, and resource spine.
MendSpeech Complete 56 Day Plan	MendSpeech_Complete_56_Day_Plan.pdf	All daily tasks in one document.
Week 1: Audio, Degradation, and Measurement Foundations	Week_1_MendSpeech_Daily_Plan.pdf	Build the audio laboratory and make SpeechDamageBench a deterministic standalone package.
Week 2: ASR, CTC, Confidence, and Repair Localization	Week_2_MendSpeech_Daily_Plan.pdf	Build the recognition and uncertainty layer, plus a reusable Modal execution path.
Week 3: Conformer From First Principles	Week_3_MendSpeech_Daily_Plan.pdf	Implement the core encoder pieces so model behavior is not a black box.
Week 4: FastConformer and Efficient Encoder Behavior	Week_4_MendSpeech_Daily_Plan.pdf	Measure why FastConformer is efficient and freeze a reproducible baseline.
Week 5: Streaming, Cache Aware Inference, and Adaptive Context	Week_5_MendSpeech_Daily_Plan.pdf	Turn the recognizer into a real time system and test uncertainty guided context spending.
Week 6: Robustness, Fine Tuning, RNNT, and Calibration	Week_6_MendSpeech_Daily_Plan.pdf	Adapt the recognizer to damaged speech while learning training and calibration discipline.
Week 7: TTS, Speaker Preservation, and Boundary Matched Reconstruction	Week_7_MendSpeech_Daily_Plan.pdf	Build MendSpeech V1 as a cascaded selective repair baseline with explicit seam diagnostics.
Week 8: Research Capstone: Cascaded Versus Direct Repair	Week_8_MendSpeech_Daily_Plan.pdf	Freeze the benchmark, run controlled ablations, compare architectures, and publish a reproducible result.

Recommended Order

1. Read the Project Blueprint once before starting.
2. Read the Master Roadmap to understand workload and architecture.
3. Use one weekly PDF at a time during execution.
4. Use the Complete 56 Day Plan as the searchable reference copy.
5. In Week 8, return to the Blueprint and freeze only claims that the evidence supports.